Lecture 7 Price discrimination

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Lecture 7 Outline

- Incentives and conditions for price discrimination
- First degree Price discrimination
- Second degree Price discrimination
  - One two-part tariff
  - Two two-part tariff
- Third degree Price discrimination
- Extra Reading:
  - Carlton and Perloff (1999) Modern industrial organization. Chapter 9-10
  - Lipczynski, Wilson and Goddard (2005). Industrial organization: competition, strategy, policy. Pearson Education. Chapter 14.3

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Some questions to answer

- Why monopolist like price discrimination?
- What are the necessary conditions for price discrimination to occur?
- What are the common types of non-uniform pricing?
- What are the welfare effects of price discrimination?

Incentives and conditions for price discrimination

- Incentive: remember a monopolist's trade-off?
- Conditions: to make price discrimination possible:
  - A firm must have some market power.
  - The firm must know or able to infer consumers' (distribution of) willingness to pay: identification problem.
  - A firm must be able to prevent or limit resales by costumers. If the group charged the lower price can resell to the other groups at a lower price than the firm charges them, no one in the latter group would buy directly from the firm.

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Crimping

- Companies with market power occasionally engage in intentional quality reduction of a portion of their output as a means of offering two qualities of goods for the purpose of price discrimination, even absent a cost saving.
- Deneckere and McAffee (1996) argue that crimping has been a frequent practice of manufacturers throughout history.
- Have a familiar feeling?

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How to prevent resales?

- Services: Most services cannot be resold because they have be consumed at the same time they are produced. For example, a dentist's treatment, an art show, an IE IV class (or is it?)
- Warranties: A manufacturer can void a warranty if a product is resold. For example, many electrical cars provide life-time services only to the initial buyer.
- Adulteration: A manufacturer can adulterate a product to make it unfit for other uses. For example, alcoholic beverages and medical alcohol. It is common to add ingredients in the latter to prevent resale. (Is this a two-way thing?)



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How to prevent resales?

- Transaction costs: If consumers has to spend money or time in reselling, resales will be more difficult.
- Contractual remedies/Government intervention: Forbids resale as part of its term of sale.
- Vertical integration: A special type of price discrimination happening between upstream firms and downstream business: if you are worried that the resale between the downstream businesses, buy!



The three types of price discrimination

Price discrimination is a pricing strategy where a firm charges different prices to different customers for the same product or service, based on various factors such as willingness to pay, quantity consumed, or customer characteristics. Price discrimination can be classified into three degrees:

1. First-degree price discrimination (also known as personalized pricing or perfect price discrimination): In this form of price discrimination, the firm charges each customer the maximum price they are willing to pay for the product or service. This allows the firm to capture the entire consumer surplus and convert it into producer surplus. First-degree price discrimination requires the firm to have detailed information about each customer's willingness to pay, which can be challenging to obtain in practice. Examples include personalized pricing in online retail based on browsing history, income level, or location, and negotiated prices in car sales or real estate transactions.
2. Second-degree price discrimination: This type of price discrimination occurs when a firm charges different prices based on the quantity consumed or the version of the product or service. The firm does not have perfect information about each consumer's willingness to pay, so it offers various pricing schemes to capture consumer surplus across different market segments. Customers self-select into the appropriate pricing scheme based on their preferences or willingness to pay. Examples include volume discounts, non-linear pricing, product versioning (e.g., basic, premium, and deluxe packages), and bundling.
3. Third-degree price discrimination: In third-degree price discrimination, the firm charges different prices to different customer groups based on observable characteristics such as age, location, or income level. The firm segments the market based on these characteristics and sets different prices for each segment, assuming that each segment has a different price elasticity of demand. Examples include student or senior citizen discounts, regional pricing differences, and different prices for business and leisure travelers in the airline industry.

All forms of price discrimination aim to increase the firm's profits by capturing more consumer surplus and adjusting prices to better match customers' willingness to pay. However, the degree of price discrimination and the information required to implement each strategy varies.

Regenerate response

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First degree Price discrimination

Welfare analysis

- Compare perfect competition, monopoly (with uniform price), and monopoly with first degree price discrimination. What can you learn from the welfare analysis?

	Consumer welfare	Producer welfare	Total welfare
Perfect competition			
Monopoly with uniform price			
Monopoly with perfect price discrimination			



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First degree Price discrimination

Fairness and efficiency

- Fairness often comes with a price.
- The Way of natural is to reduce excess and supplement insufficiency; the way of human is otherwise, to reduce the deficient and strive for the excessive — Tao Te Ching (Lao Tzu), translated by Chatgpt

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Second degree Price discrimination

Non-linear pricing

- The firm charges different prices based on the quantity the costumer buys. It allows a company to increase profits without knowing consumers' exact willingness to pay.

Two-part tariffs

- It consists of a lump sum payment P_{ls} for a good or service combined with a per unit user charge P_u.

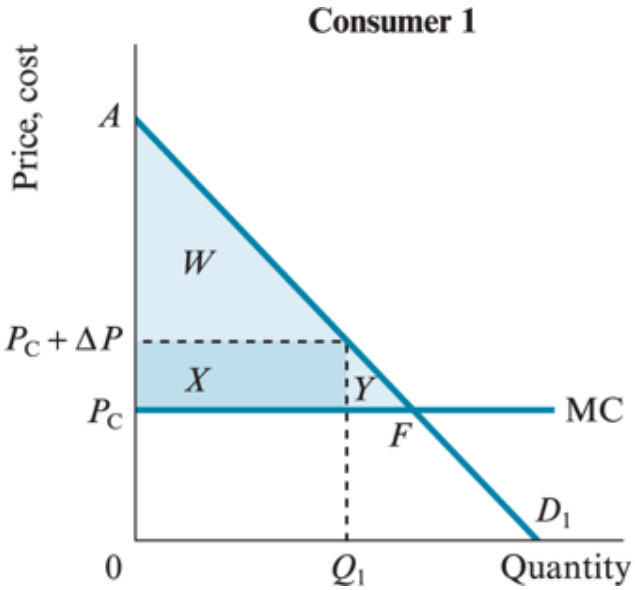
Buffet

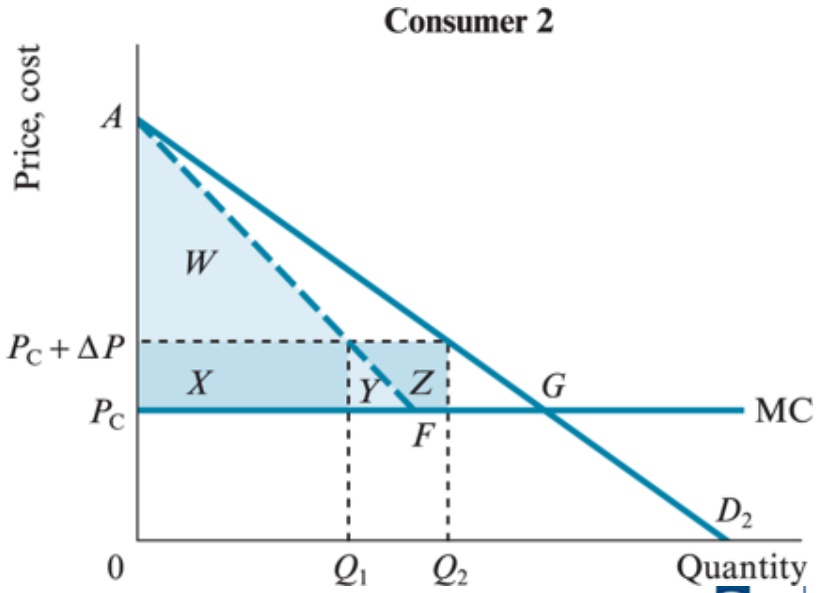
- The most classical two-part tariff, with a positive P_{ls} as the entry fee and a zero P_u as "eat as much as you want".

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One two-part tariff

- At the same unit price, consumer 2 buys more than consumer 1. The dashed curve in the right graph is consumer 1's demand.
- The firm has following options of two-part tariffs.
  - Option 1: $P_u = P_c$ and $P_{ls} = P_c AF$
  - Option 2: $P_u = P_c$ and $P_{ls} = P_c AG$
  - Option 3: $P_u = P_c + \Delta P$ and $P_{ls} = W$





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One two-part tariff

- The first two strategies are common. The better from those two depends on whether $2P_cAF > P_cAG$, which is valid as shown in the graph. Hence, option 1 is better and the firm's profit is $2(W + X + Y)$. However, it is still not optimal for the firm.
- Actually, there is still room for improvement. When option 3 is used, both consumers pay the lump-sum price and each buys Q_1 and Q_2, respectively. The firm earns profit $W + X$ from consumer 1, $W + X + Y + Z$ from consumer 2. In total, the firm's profit becomes $2(W + X) + Y + Z$. Since $Z > Y$ as shown in the graph, option 3 beats option 1.



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Let us get skeptical:

- Is it possible that option 3 is worse than option 1?
- Is it possible to set the unit price lower than the marginal cost? If yes, any example for that?
- What happens if we have three groups?



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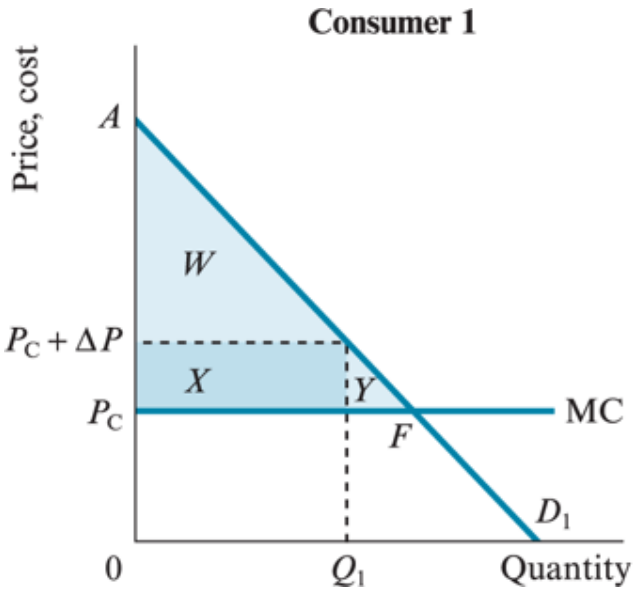
Two two-part tariff

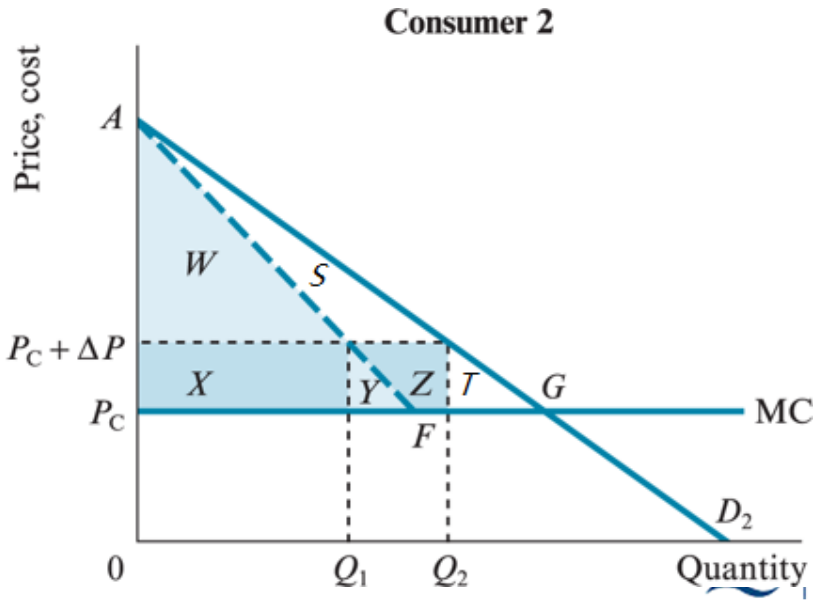
- Now, if the firm could charge two separate two-part tariffs, is it possible for the firm to earn a higher profit?
- Since the firm cannot identify between two groups, the key is to let the two groups spontaneously choose the different options. Otherwise, it would be equivalent to one two-part tariff.
- We can easily imagine that, across the two two-part tariffs, one has the lower P_{ls} and the other has the lower P_u.
- Now suppose we have an option 4 in additional to option 3. Consumers could freely choose between the these two options.
  - Option 3: $P_u = P_c + \Delta P$ and $P_{ls} = W$
  - Option 4: $P_u = P_c$ and $P_{ls} = W + X + Y + Z + \epsilon$. Let's take $\epsilon = \frac{T}{2}$ to start with.

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Two two-part tariff

- Clearly consumer 1 will not choose option 4.
- Consumer 2's consumer surplus is S when choosing option 3, and $S + T - \epsilon$ when choosing option 4. A simple choice.
- The firm earns a profit of $W + X$ from consumer 1, $W + X + Y + Z + \epsilon$ from consumer 2. The total profit increases by ϵ comparing to when he offers one two-part tariff option 3. (so what should be the ϵ?)





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Two two-part tariff

- About **incentive compatible** and **individual rationality** ...
- Any example of the two two-part tariffs?



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Third degree Price discrimination

How it works

- Monopoly in different groups.

$$max\pi = \sum_i (p_i - c)q_i(p_i)$$

$$\Longrightarrow \frac{\partial \pi}{\partial p_i} = q_i(p_i) + (p_i - c)\frac{\partial q_i}{\partial p_i} = 0$$

$$\Longrightarrow L_i = \frac{p_i - c}{p_i} = -\frac{q_i(p_i)}{p_i}\frac{\partial p_i}{\partial q_i(p_i)} = -\frac{\frac{\partial p_i}{p_i}}{\frac{\partial q_i(p_i)}{q_i(p_i)}} = \frac{1}{\eta_i}$$



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For example

Here are some examples of how a movie theatre can use third-degree price discrimination to maximize profit:

1. Segmenting Customers by Time: A movie theatre may charge different prices for different showtimes to appeal to different customer segments. For instance, the theatre may charge a lower price for matinee showtimes to attract price-sensitive customers, and a higher price for evening showtimes to capture more of the consumer surplus from those willing to pay more. This strategy can also help the theatre to fill seats during slower times of the day.
2. Segmenting Customers by Age: A movie theatre may charge different prices for different age groups. For instance, the theatre may charge a lower price for children and seniors, who are typically more price-sensitive, and a higher price for adults who are willing to pay more. This strategy can help the theatre to attract more customers across different age groups and capture more of the consumer surplus from each group.
3. Segmenting Customers by Membership: A movie theatre may offer different prices for members and non-members. For instance, the theatre may offer a lower price for members who pay an annual fee and have access to exclusive discounts, rewards, and benefits. This strategy can help the theatre to retain loyal customers and generate additional revenue from membership fees.
4. Segmenting Customers by Location: A movie theatre may charge different prices for different locations, such as urban vs. suburban areas. For instance, the theatre may charge a higher price for urban locations where customers may have a higher willingness-to-pay, and a lower price for suburban locations where customers may be more price-sensitive. This strategy can help the theatre to tailor prices to the local market conditions and maximize revenue in each location.



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For example

Here are some examples of how an airline can use third-degree price discrimination to maximize profit:

1. Segmenting Customers by Time of Booking: Airlines may offer different prices for tickets booked well in advance or at the last minute. Customers who book well in advance may be charged a lower price, while customers who book at the last minute may be charged a higher price due to the urgency of their travel needs.
2. Segmenting Customers by Destination: Airlines may offer different prices for different destinations based on their popularity and demand. For instance, the airline may charge a higher price for popular tourist destinations during peak travel season and a lower price for less popular destinations or during off-peak travel season.
3. Segmenting Customers by Seat Class: Airlines may charge different prices for different seat classes, such as economy, business, or first class. The airline may charge a higher price for premium seat classes to capture more of the consumer surplus from customers who are willing to pay more for extra comfort and amenities.
4. Segmenting Customers by Age: Airlines may offer different prices for children, students, or seniors who may be more price-sensitive than other customers. For instance, the airline may offer a discounted price for children or students who are traveling for educational purposes.
5. Segmenting Customers by Loyalty: Airlines may offer different prices for loyal customers who have earned rewards points or frequent flyer miles. These customers may be offered discounts or exclusive prices for tickets, which can help to retain their loyalty and generate additional revenue from rewards programs.



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Screening

- Notice that it is possible for a firm to use third-degree price discrimination even if it cannot clearly identify different groups of consumers.
- In that case, a firm can use proxy variables to infer differences in willingness-to-pay and segment customers accordingly: this is called screening.
- The basic intuition is that, elastic costumers will spend time/energy in seeking for cheaper products and inelastic costumers will not.



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Screening


bp
SPAR
regular unleaded 140.9
regular diesel 147.9
autogas 81.9
ultimate
CUT PRICE FUEL
2 DAYS A WEEK

PD or PD?

- Although very distinctive in the definition, it could be sometimes tricky to identify a strategy to be product differentiation or price differentiation.
- For example: hardcover versus softcover textbooks, business vs economy class airline tickets, and so on. Are those products homogenous or not?
- Several factors: the costs to provide, the necessity to ban the resale...

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Thank you



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